

Supplementary Material

DEM-derived pore reconstruction and debris-transport assessment in a Li_4SiO_4 pebble bed

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Scope of the Supplementary Material

This Supplementary Material provides supporting figures and table descriptors for the DEM-derived pore reconstruction, debris-transport analysis, hydraulic-response checks and reproducibility records discussed in the main manuscript. The numerical source tables are provided as comma-separated files to keep the quantitative evidence traceable and reusable.

Supplementary Figure Roadmap

Figure	Role	Evidence boundary
Fig. S1	time-resolved deposition and BTC for representative loading case	Representative time-resolved support for the delayed leading-tail interpretation.
Fig. S2	front-migration speed fits	Front-speed estimates for the representative drive case.
Fig. S3	voxel-network pressure pilot	Scalar voxel-network pressure-response check.
Fig. S4	conductance-exponent sensitivity for cropped peak-blockage domains	Sensitivity of the scalar conductance response to the conductance exponent.
Fig. S5	near-threshold repeat-seed stochastic variability	Repeat-seed variability for the near-threshold DEM setting.
Fig. S6	returned cropped-domain OpenFOAM pressure evidence for peak-blockage handoff cases	Cropped-domain OpenFOAM pressure-response check.
Fig. S7	ParaView-rendered OpenFOAM cropped-domain model and mesh slice	Representative cropped-domain OpenFOAM geometry and mesh visualization.
Fig. S8	spatial heterogeneity data mining of axial deposition profiles, localization relaxation and downstream spreading	Spatial-deposition heterogeneity metrics derived from the axial blockage profiles.

Supplementary Figures

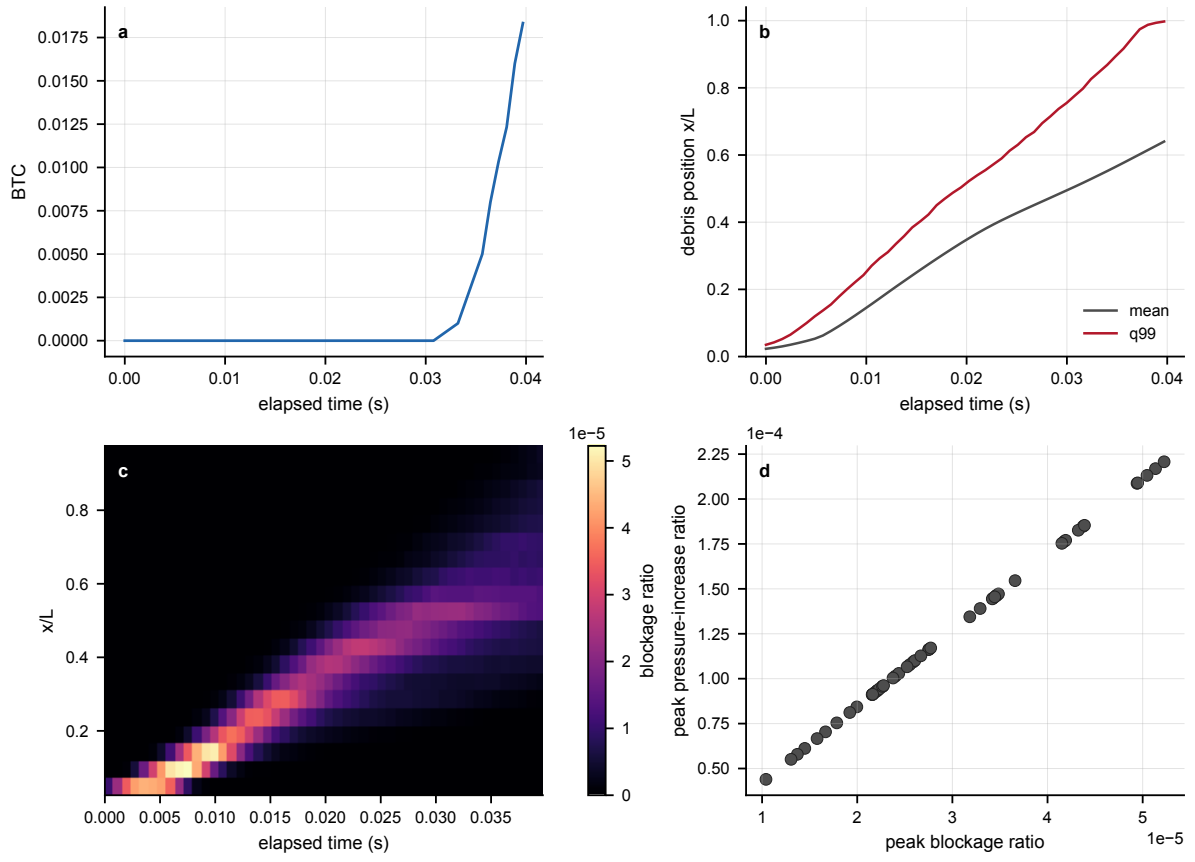


Fig. S1. time-resolved deposition and BTC for representative loading case. *Boundary:* Representative time-resolved support for the delayed leading-tail interpretation.

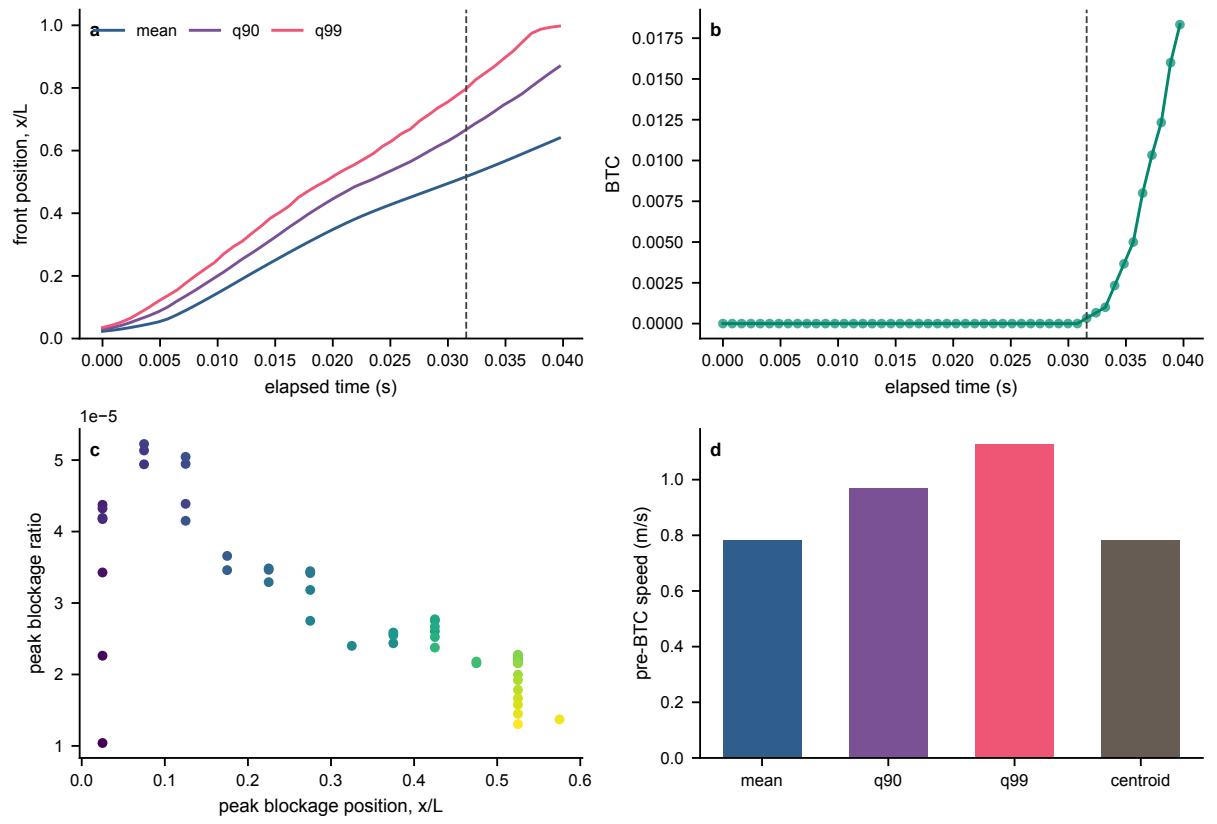


Fig. S2. front-migration speed fits. *Boundary:* Front-speed estimates for the representative drive case.

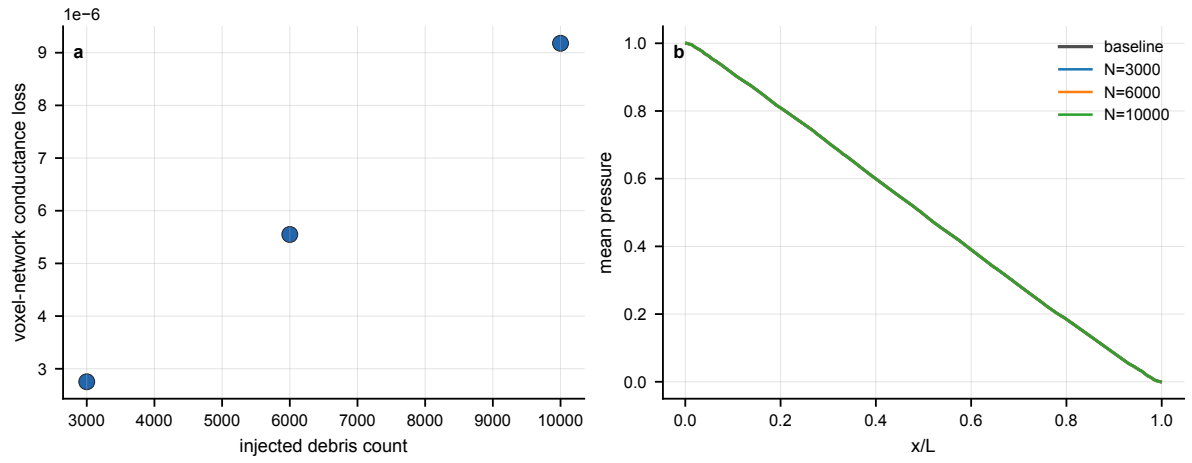


Fig. S3. voxel-network pressure pilot. *Boundary:* Scalar voxel-network pressure-response check.

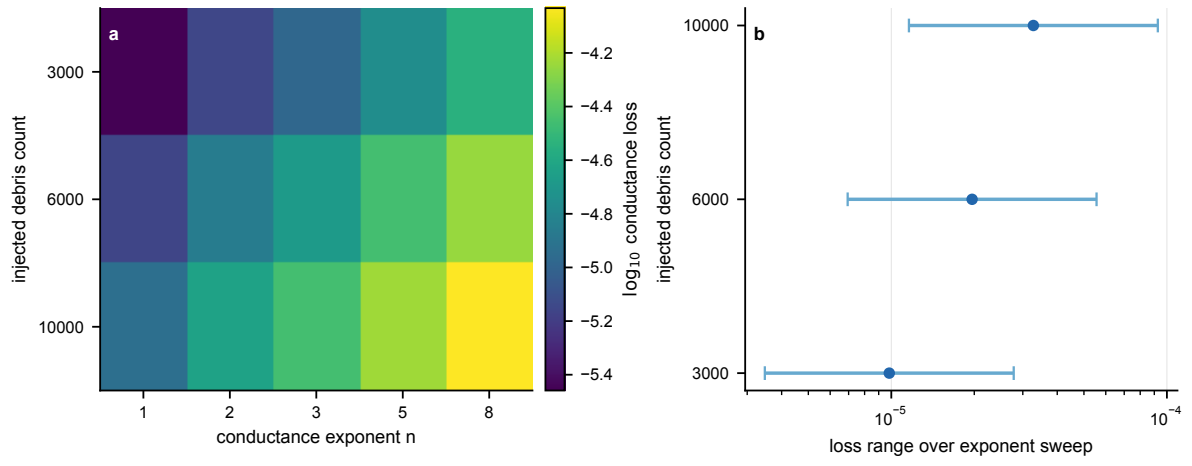


Fig. S4. conductance-exponent sensitivity for cropped peak-blockage domains. *Boundary:* Sensitivity of the scalar conductance response to the conductance exponent.

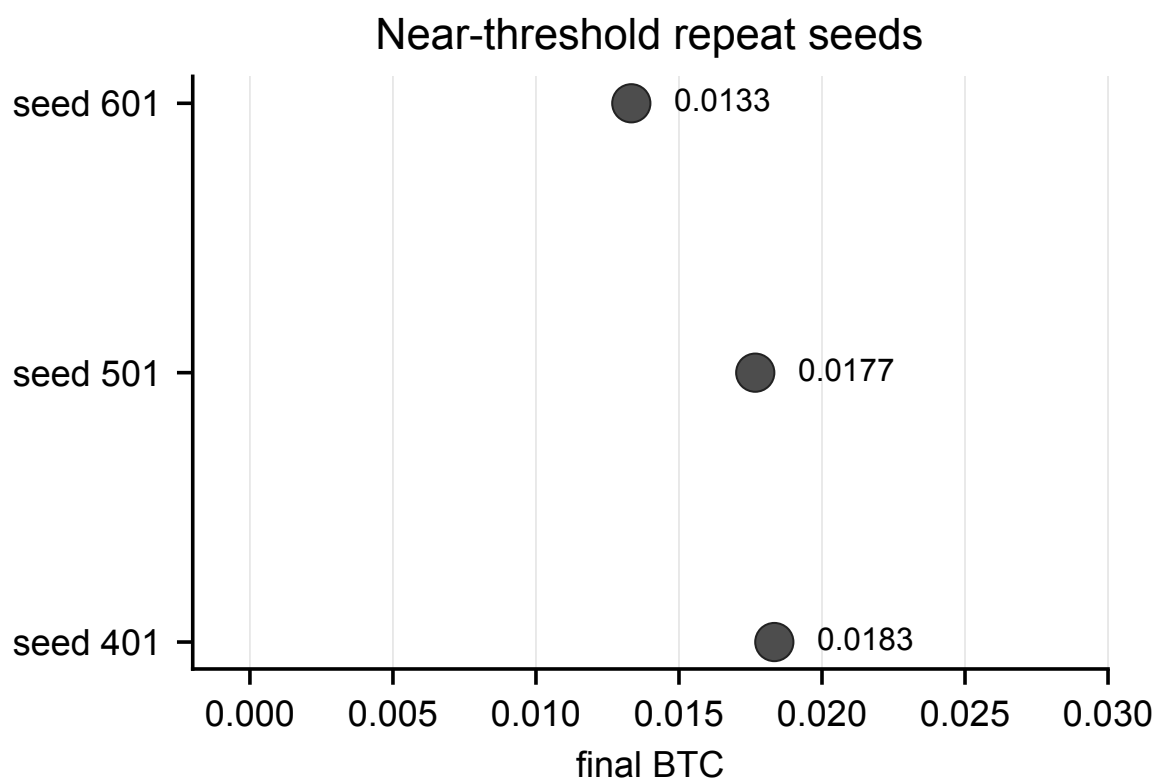


Fig. S5. near-threshold repeat-seed stochastic variability. *Boundary:* Repeat-seed variability for the near-threshold DEM setting.

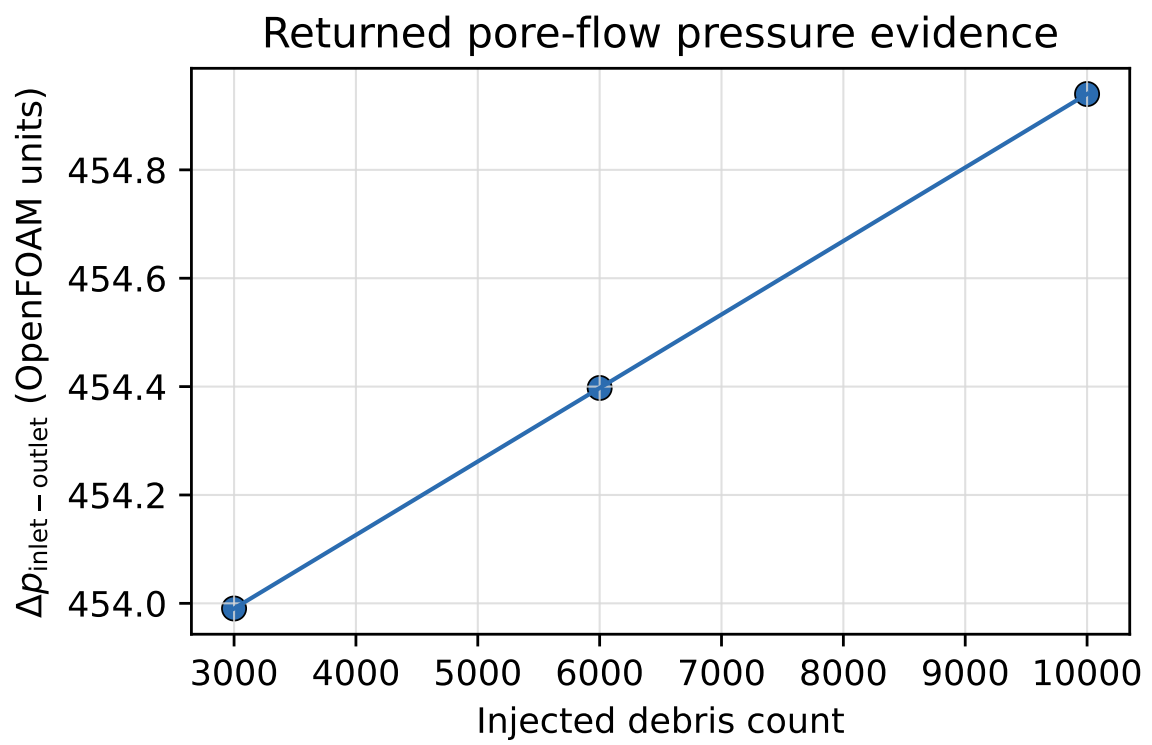


Fig. S6. returned cropped-domain OpenFOAM pressure evidence for peak-blockage handoff cases. *Boundary:* Cropped-domain OpenFOAM pressure-response check.

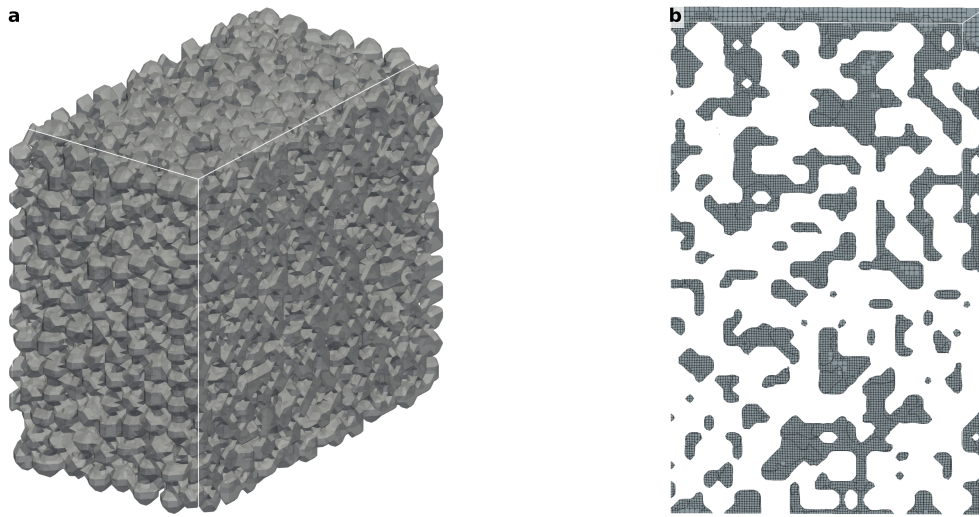


Fig. S7. ParaView-rendered OpenFOAM cropped-domain model and mesh slice. *Boundary:* Representative cropped-domain OpenFOAM geometry and mesh visualization.

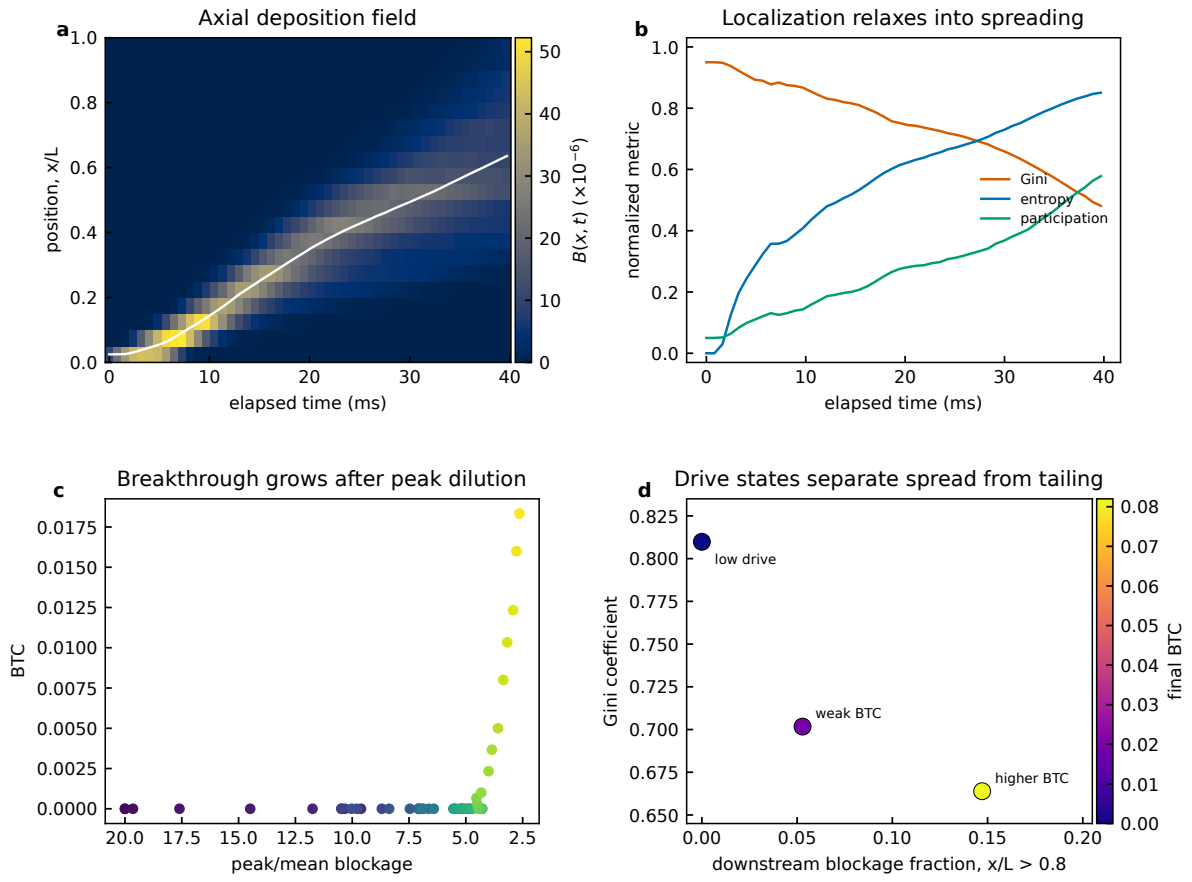


Fig. S8. spatial heterogeneity data mining of axial deposition profiles, localization relaxation and downstream spreading. *Boundary:* Spatial-deposition heterogeneity metrics derived from the axial blockage profiles.

Supplementary Table Roadmap

The numerical source tables are provided as comma-separated files to keep the quantitative evidence machine-readable. The table below records their role and column structure.

Table	Role	File	Columns
Table S1	table_s1_case_matrix	table_s1_case_matrix.csv	table_group, case_id, evidence_level, df_over_dp, gas_velocity_m_s, ...
Table S2	table_s2_structural_metrics	table_s2_structural_metrics.csv	case_group, case_id, outlet_connected_void_fraction, largest_connected_void_fraction, topological_charge, ...
Table S3	table_s3_resolution_and_status	table_s3_resolution_and_status.csv	group, item, status, metric_1, metric_2, ...
Table S4	table_s4_mechanism_evidence_matrix	table_s4_mechanism_evidence_matrix.csv	mechanism, evidence, metric, value, interpretation, ...
Table S5	table_s5_literature_positioning_matrix	table_s5_literature_positioning_matrix.csv	reference_key, study_context, method_family, tracks_fine_particles, reports_breakthrough_or_retention, ...
Table S6	table_s6_pressure_proxy_summary	table_s6_pressure_proxy_summary.csv	bris_total_number, gas_velocity_m_s, baseline_pressure_gradient_pa_m, profile_mean_pressure_gradient_pa_m, peak_local_pressure_gradient_pa_m, ...
Table S7	table_s7_voxel_pressure_pilot_summary	table_s7_voxel_pressure_pilot_summary.csv	case_label, debris_total_number, pressure_model, status, through_pore_voxels, ...
Table S8	table_s8_cropped_flow_domain_manifest	table_s8_cropped_flow_domain_manifest.csv	case_label, debris_total_number, crop_role, x_peak_blockage_m, x_applied_lower_m, ...
Table S9	table_s9_cropped_flow_permeability_summary	table_s9_cropped_flow_permeability_summary.csv	case_label, debris_total_number, pressure_model, baseline_conductance, loaded_conductance, ...
Table S10	table_s10_cropped_flow_permeability_sensitivity	table_s10_cropped_flow_permeability_sensitivity.csv	case_label, debris_total_number, conductance_mapping, conductance_exponent, relative_resistance, ...
Table S11	table_s11_benchmark_crosswalk	table_s11_benchmark_crosswalk.csv	benchmark_id, reviewer_question, comparable_metric, current_numeric_evidence, comparison_type, ...
Table S12	table_s12_dimensionless_mechanism_map	table_s12_dimensionless_mechanism_map.csv	mechanism_family, case_label, df_over_dp, gas_velocity_m_s, drag_to_weight_ratio, ...

Table	Role	File	Columns
Table S13	table_s13_mechanism_claim_table	table_s13_mechanism_claim_table.csv	claim_id, mechanism_claim, primary_coordinate, evidence_rows, coordinate_range, ...
Table S14	table_s14_near_threshold_repeat_seed_summary	table_s14_near_threshold_repeat_seed_summary.csv	focusal_processed_count, final_BTC_mean, final_BTC_std, final_BTC_min, final_BTC_max, ...
Table S15	table_s15_3d_resistance_amplification_sensitivity	table_s15_3d_resistance_amplification_sensitivity.csv	cases_insensitivity, bris_total_number, resistance_mapping, amplification_factor, relative_resistance, ...
Table S16	table_s16_reviewer_evidence_closure_matrix	table_s16_reviewer_evidence_closure_matrix.csv	risk_and_severity, closure_state, linked_claims, linked_benchmarks, ...
Table S17	table_s17_mechanism_state_classifier	table_s17_mechanism_state_classifier.csv	evidence_family, case_label, final_BTC, retention, source_fraction, ...
Table S18	table_s18_drag_scaling_summary	table_s18_drag_scaling_summary.csv	evidence_family, case_label, df_over_dp, gas_velocity_m_s, debris_diameter_um, ...
Table S19	table_s19_pressure_evidence_ladder	table_s19_pressure_evidence_ladder.csv	level, evidence_class, model_or_data, case_count, response_metric, ...

Boundary Statement

All supplementary plots and tables should be interpreted within the studied parameter space, the present one-way DEM drag assumptions, finite voxel resolution and the absence of resolved hydraulic feedback in the transport calculation. Breakthrough curves, local blockage, connected-pore descriptors and pressure-drop checks are deliberately separated rather than collapsed into a single predictive clogging criterion.